CMPE403 DATA SCIENCE AND ANALYTICS PROJECT

FINAL PROJECT REPORT

Analyzing the Impact of Academic and Lifestyle Factors on Student Depression Using Classification Algorithms

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1 INTRODUCTION

Depression is a health problem that significantly reduces a person's quality of life [1]. In recent years, students have been the group most affected by depression [2]. The stress of exams and the desire to succeed, experienced during the intensity of academic life, create immense psychological pressure on students. However, academic life is not the only factor contributing to depression in students. External factors such as lifestyle and economic conditions also greatly influence depression. A student who is depressed both has a lower quality of life and finds it very difficult to achieve success in their academic life.

Depression is a very complex biological process and can be influenced by many different factors. This makes analyzing the factors that cause depression quite difficult. A high-quality depression analysis requires the simultaneous evaluation of variables such as academic, demographic, and lifestyle factors. Data science and machine learning methods can yield highly meaningful results from very large datasets [3]. In particular, using machine learning models, it is possible to obtain very reliable results.

This project uses a real dataset containing numerous variables related to students' academic, demographic, and lifestyle characteristics to analyze the factors influencing their depression status. The main goal of this project is to analyze whether students are depressed using the variables in the dataset and the performance of different machine learning models.

This project aims to better analyze the relationship between students and depression and contribute to the early diagnosis of students at risk. The aim is also to raise public awareness that methods like machine learning have many significant impacts on the healthcare field.

In addition, this project involves many processes such as data understanding, preprocessing, model development, and analysis. Within these processes, it aims to understand whether a student is depressed and to what extent different values are effective in determining depression by comparing different classification-based machine learning models.

2 METHODOLOGY

2.1 Dataset Description

This project utilized the Student Depression Dataset which was found on Kaggle [4], and machine learning models were used to analyze the extent to which various factors influence students' depression levels. This dataset contains 27,901 observations that help explain the relationship between students and depression. Academic life, with its intensity and, most importantly, the stress it causes, is the primary factor influencing students' depression. However, depression is a complex biological process, and many factors outside of academic life significantly affect students' depression. Therefore, variables such as demographic characteristics and lifestyle are also included in the dataset.

The main goal in using this dataset is to analyze the student's depression status. Therefore, the target variable is "Depression". According to this dataset, the student is either depressed or not. For this reason, the "Depression" variable is binary; if the student is not depressed, the value is 0, and if they are depressed, the value is 1. Machine learning models were used to accurately analyze both the factors associated with the student's depression and the target variable.

The dataset includes academic variables such as CGPA, Academic Pressure, and Study Satisfaction, as well as demographic characteristics such as age, gender, and city of residence. The presence of multiple categorical factors makes the analyses much more powerful. However, these factors alone are insufficient to analyze a complex biological process like depression. Therefore, factors such as Sleep Duration, Dietary Habits, Work/Study Hours, Work Pressure, and Job Satisfaction, which are thought to significantly influence psychology and reflect students' lifestyles, are also included in the dataset.

The dataset also included factors such as Financial Stress, Family History of Mental Illness, and whether students have ever had suicidal thoughts, all of which are thought to have a greater influence on whether a student is depressed compared to other variables. All these variables are sufficient to reliably analyze whether a student is depressed.

Thanks to the powerful variables contained within the selected dataset, factors that either cause or have a significant impact on students' depression were easily analyzed. Furthermore, these variables make the dataset highly suitable for classification-based machine learning models.

2.2 Data Preprocessing

Before we applied any machine learning algorithms, we had to examine the dataset and its quality. The dataset was carefully inspected for any missing or duplicate values, and invalid entries. Data preprocessing was one of the critical steps in this project as noisy or inconsistent data can negatively impact model performance and lead to unreliable predictions.

First, we examined the structure of the dataset to understand variable types. In this step, we looked into feature types, number of entries and number of non-null values. As seen in Figure 2.1, there were 27901 rows in the dataset and the feature types varied between int64, object, and float64.

```
[72]: import pandas as pd
      import numpy as np

      df = pd.read_csv("student_depression_dataset.csv")

      df.info()
      df.head()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27901 entries, 0 to 27900
Data columns (total 18 columns):
 #   Column                                 Non-Null Count  Dtype
---  ------                                 --------------  -----
 0   id                                     27901 non-null  int64
 1   Gender                                 27901 non-null  object
 2   Age                                    27901 non-null  float64
 3   City                                   27901 non-null  object
 4   Profession                             27901 non-null  object
 5   Academic Pressure                      27901 non-null  float64
 6   Work Pressure                          27901 non-null  float64
 7   CGPA                                   27901 non-null  float64
 8   Study Satisfaction                     27901 non-null  float64
 9   Job Satisfaction                       27901 non-null  float64
 10  Sleep Duration                         27901 non-null  object
 11  Dietary Habits                         27901 non-null  object
 12  Degree                                 27901 non-null  object
 13  Have you ever had suicidal thoughts ?  27901 non-null  object
 14  Work/Study Hours                       27901 non-null  float64
 15  Financial Stress                       27901 non-null  object
 16  Family History of Mental Illness       27901 non-null  object
 17  Depression                             27901 non-null  int64
dtypes: float64(7), int64(2), object(9)
memory usage: 3.8+ MB
```

Figure 2.1 Feature types

After this step, we quantified missing values for each feature to determine whether imputation or removal of any row was necessary. As seen in Figure 2.2, there was no missing values.

```
[73]: missing_values = df.isnull().sum()
      print(missing_values)

      missing_percent = (missing_values / len(df)) *100
      print(missing_percent)

id                                       0
Gender                                   0
Age                                      0
City                                     0
Profession                               0
Academic Pressure                        0
Work Pressure                            0
CGPA                                     0
Study Satisfaction                       0
Job Satisfaction                         0
Sleep Duration                           0
Dietary Habits                           0
Degree                                   0
Have you ever had suicidal thoughts ?    0
Work/Study Hours                         0
Financial Stress                         0
Family History of Mental Illness         0
Depression                               0
dtype: int64
id                                       0.0
Gender                                   0.0
Age                                      0.0
City                                     0.0
Profession                               0.0
Academic Pressure                        0.0
Work Pressure                            0.0
CGPA                                     0.0
Study Satisfaction                       0.0
Job Satisfaction                         0.0
Sleep Duration                           0.0
Dietary Habits                           0.0
Degree                                   0.0
Have you ever had suicidal thoughts ?    0.0
Work/Study Hours                         0.0
Financial Stress                         0.0
Family History of Mental Illness         0.0
Depression                               0.0
dtype: float64
```

Figure 2.2 Count of missing values

Duplicate values can bias classification models by over-representing certain response patterns. Therefore, we examined to see whether there were any duplicate rows. As seen in Figure 2.3, there were not any duplicate rows.

```
[74]: num_duplicates = df.duplicated().sum()
      print(f"Number of duplicate rows: {num_duplicates}")

      df = df.drop_duplicates()

Number of duplicate rows: 0
```

Figure 2.3 Count of duplicate rows

In the next step, we checked the numerical variables against expected ranges based on the dataset description. For instance, Likert-scale variables were restricted to their valid minimum and maximum ranges as seen in Figure 2.4. There were not any invalid values found.

```
[75]: expected_ranges = {
          'Academic Pressure': (0, 5),
          'Work Pressure': (0, 5),
          'CGPA': (0,10),
          'Study Satisfaction': (0, 5),
          'Job Satisfaction': (0, 5),
          'Work/Study Hours': (0, 12),
          'Depression': (0, 1),
      }
```

```
[76]: for col, (min_val, max_val) in expected_ranges.items():
          if col in df.columns:
              invalid = df[(df[col] < min_val) | (df[col] > max_val)]
              if invalid.empty:
                  print(f"{col}: no invalid values found.")
              else:
                  print(f"{col}: invalid values detected.")
```

```
Academic Pressure: no invalid values found.
Work Pressure: no invalid values found.
CGPA: no invalid values found.
Study Satisfaction: no invalid values found.
Job Satisfaction: no invalid values found.
Work/Study Hours: no invalid values found.
Depression: no invalid values found.
```

Figure 2.4 Variables checked against ranges

The dataset contained several categorical variables such as gender, dietary habits, sleep duration, profession, financial stress and city. Since most machine learning algorithms require numerical input, we transformed these variables into numerical values using label encoding. Binary categorical variables such as family history of mental illness and suicidal thoughts were encoded as 0 (No) and 1(Yes) as shown in Figure 2.5.

```
import numpy as np
import json
from sklearn.preprocessing import LabelEncoder

category_mappings = {}
binary_map = {'Yes': 1, 'No': 0}

binary_cols = [
    'Have you ever had suicidal thoughts ?',
    'Family History of Mental Illness'
]

for col in binary_cols:
    df[col] = df[col].map(binary_map)
    category_mappings[col] = binary_map
```

Figure 2.5 Label encoding binary variables

For ordinal values, we encoded them in a way that would preserve their ranking since they have a natural order. As seen in Figure 2.6, sleep duration and dietary habits were encoded to reflect their increasing levels of quality or quantity.

```
df['Sleep Duration'] = df['Sleep Duration'].replace('Unknown', np.nan)
df['Dietary Habits'] = df['Dietary Habits'].replace('Unknown', np.nan)

sleep_map = {
    'Less than 5 hours': 0,
    '5-6 hours': 1,
    '7-8 hours': 2,
    'More than 8 hours': 3
}

diet_map = {
    'Unhealthy': 0,
    'Moderate': 1,
    'Healthy': 2
}

df['Sleep Duration'] = df['Sleep Duration'].map(sleep_map)
df['Dietary Habits'] = df['Dietary Habits'].map(diet_map)

category_mappings['Sleep Duration'] = sleep_map
category_mappings['Dietary Habits'] = diet_map


df = df.dropna()
```

Figure 2.6 Encoding ordinal variables

For nominal variables such as city, profession, degree and gender we encoded them using label encoding which assigned a unique integer to each category as shown in Figure 2.7.

```
le = LabelEncoder()
categorical_cols = ['Gender', 'City', 'Profession', 'Degree']

for col in categorical_cols:
    df[col] = le.fit_transform(df[col])
    mapping_dict = dict(zip(le.classes_.astype(str), range(len(le.classes_))))
    category_mappings[col] = mapping_dict
```

Figure 2.7 Encoding nominal variables

As seen in Figure 2.8, we have saved the mappings into a json file and saved the encoded and cleaned dataset into a new csv file. Finally, we checked all the features to ensure that everything was correct.

```
]: with open('category_mappings_v2.json', 'w') as f:
       json.dump(category_mappings, f, indent=4)

   df.to_csv('processed_data_v2.csv', index=False)

   print("Data cleaned, encoded, and saved.")
   print("\nFinal Data Types:")
   print(df.dtypes)
```

```
Data cleaned, encoded, and saved.

Final Data Types:
Gender                                    int64
Age                                     float64
City                                      int64
Profession                                int64
Academic Pressure                       float64
Work Pressure                           float64
CGPA                                    float64
Study Satisfaction                      float64
Job Satisfaction                        float64
Sleep Duration                          float64
Dietary Habits                          float64
Degree                                    int64
Have you ever had suicidal thoughts ?     int64
Work/Study Hours                        float64
Financial Stress                        float64
Family History of Mental Illness          int64
Depression                                int64
dtype: object
```

Figure 2.8 Encoded data types

After all the steps, our dataset looked like Table 2.1 and Table 2.2 below.

Table 2.1 Cleaned and encoded dataset

ID	Gender	Age	City	Profession	Academic Pressure	Work Pressure	CGPA	Study Satisfaction	Job Satisfaction
1	1	33	31	12	5	0	8.97	2	0
2	0	24	2	12	2	0	5.90	5	0
3	1	31	25	12	3	0	7.03	5	0
4	0	28	29	12	3	0	5.59	2	0
5	0	25	11	12	4	0	8.13	3	0

Table 2.1 (Continued) Cleaned and encoded dataset

ID	Sleep Duration	Dietary Habtis	Degree	Suicidal Thoughts	Work/Study Hours	Financial Stress	Family History of Mental Ilness	Depression
1	1	1	4	1	3	1	0	1
2	1	1	11	0	3	2	1	0
3	0	0	6	0	9	1	1	0
4	2	2	8	1	4	5	1	1
5	1	1	17	1	1	1	0	0

2.3 Machine Learning Models

In this project, we have used Python and Jupyter Notebook as the programming environment to classify student depression status. Additionally, we have used libraries such as pandas, numpy, and seaborn as shown in Figure 2.9. These libraries were used to develop the models and visualize the data and results.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import GaussianNB
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, roc_auc_score
from sklearn.metrics import precision_score, recall_score, f1_score
from sklearn.metrics import classification_report, confusion matrix
```

Figure 2.9 Used libraries

We splitted the dataset into training and testing sets as seen in Figure 2.10. 80% of the dataset was used for the training set and 20% of the dataset was used for the testing set.

Additionally, we used stratified split strategy to make sure that the proportion of depressed and non-depressed students were consistent across both sets.

```
df = pd.read_csv("../Data/processed_data_v2.csv")

X = df.drop("Depression", axis=1)
y = df["Depression"]

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42, stratify=y)
```

Figure 2.10 Train-test split

Moreover, we have used three separate classification algorithms to compare their performance. These models were Logistic Regression, Naïve Bayes and Random Forest as seen in Figure 2.11.

```
logreg_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", LogisticRegression(max_iter=1000))
])
```

```
nb_pipeline = Pipeline([
    ("scaler", StandardScaler()),
    ("model", GaussianNB())
])
```

```
rf_model = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
```

Figure 2.11 Models that were used

As seen in Figure 2.12., we have evaluated the models' performances with Classification Reports which shows metrics, such as Accuracy, Precision, Recall and F1-Score.

```
models = {
    "Logistic Regression": logreg_pipeline,
    "Naive Bayes": nb_pipeline,
    "Random Forest": rf_model
}

print("Model Performance Comparison")
print("-" * 60)

for name, model in models.items():
    model.fit(X_train, y_train)

    y_pred = model.predict(X_test)

    print(f"\n{name}")
    print("-" * 40)
    print(classification_report(y_test, y_pred))
```

Figure 2.12 Code snippet for Classification Reports

We have also looked at the Confusion Matrix and Feature Importance Analysis for the Random Forest model as seen in Figure 2.13.

```
plt.figure(figsize=(6, 4))
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['No', 'Yes'], yticklabels=['No', 'Yes'])
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()

importances = rf_model.feature_importances_
feature_names = X.columns
indices = np.argsort(importances)[::-1]

plt.figure(figsize=(10, 6))
plt.title("Feature Importances (What drives Depression?)")
plt.bar(range(X.shape[1]), importances[indices], align="center")
plt.xticks(range(X.shape[1]), feature_names[indices], rotation=90)
plt.tight_layout()
plt.show()
```

Figure 2.13 Code snippet for Confusion Matrix and Feature Importance Analysis

We have also looked at he ROC-AUC curve for the model as seen in Figure 2.14.

```
from sklearn.metrics import roc_auc_score, roc_curve

y_prob = rf_model.predict_proba(X_test)[:, 1]

auc = roc_auc_score(y_test, y_prob)
print(f"ROC-AUC Score: {auc:.3f}")

fpr, tpr, _ = roc_curve(y_test, y_prob)

plt.figure(figsize=(6,4))
plt.plot(fpr, tpr, label=f'AUC = {auc:.2f}')
plt.plot([0,1], [0,1], linestyle='--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve - Random Forest')
plt.legend()
plt.show()
```

Figure 2.14 Code snippet for ROC-AUC curve

After these steps, we have also tried running the models with different parameters and initialization. We performed manual tuning on the Random Forest model, we tested with different n_estimators values of [50, 100, 200] and max_depth of [None, 10, 20]. But there was no major improvement between different values. Additionally, we have tried dropping some of the columns to see if it would affect the models' performance positively. However, there were no improvements.

3 MAIN FINDINGS

3.1 Correlation Matrix

The relationships between variables were examined using a Pearson correlation matrix as seen in Figure 3.1. The strength and direction of the relationship between the pair of variables can be determined using the coefficients obtained from this matrix. Shades closer to red indicate positive relationships, whereas shades closer to blue indicate negative relationships. Lighter colored areas represent weak, negligible relationships. In the analysis, significant increases and decreases in relationships between variables were evaluated. The focus of the study is to determine the relationships between different variables in the dataset and depression.

The correlation values between the depression variable and other variables were analyzed to identify variables showing a stronger relationship with depression. The findings indicate that stress, lifestyle, and psychosocial variables exhibit more significant relationships with depression. Specifically, stress-related variables such as academic pressure and financial stress appear to correlate positively with depression. The correlation values of 0.36 and 0.47 do not indicate a strong correlation. However, in psychosocial data, they are considered to be moderate and significant relationships and are valuable in terms of modeling. This suggests that higher stress levels increase the likelihood of depression. Conversely, lifestyle related variables such as sleep duration, dietary habits, and satisfaction levels show a negative correlation. It can be argued that these variables have a protective effect against depressive symptoms.

Correlation only captures linear relationships. Variables with weak correlations can still contribute to depression when combined with other factors. This observation further supports the use of nonlinear classification models that can capture complex feature interactions beyond linear dependencies. Therefore, information obtained from correlation analysis has been used as a preliminary step to support and interpret the results of subsequent classification models.

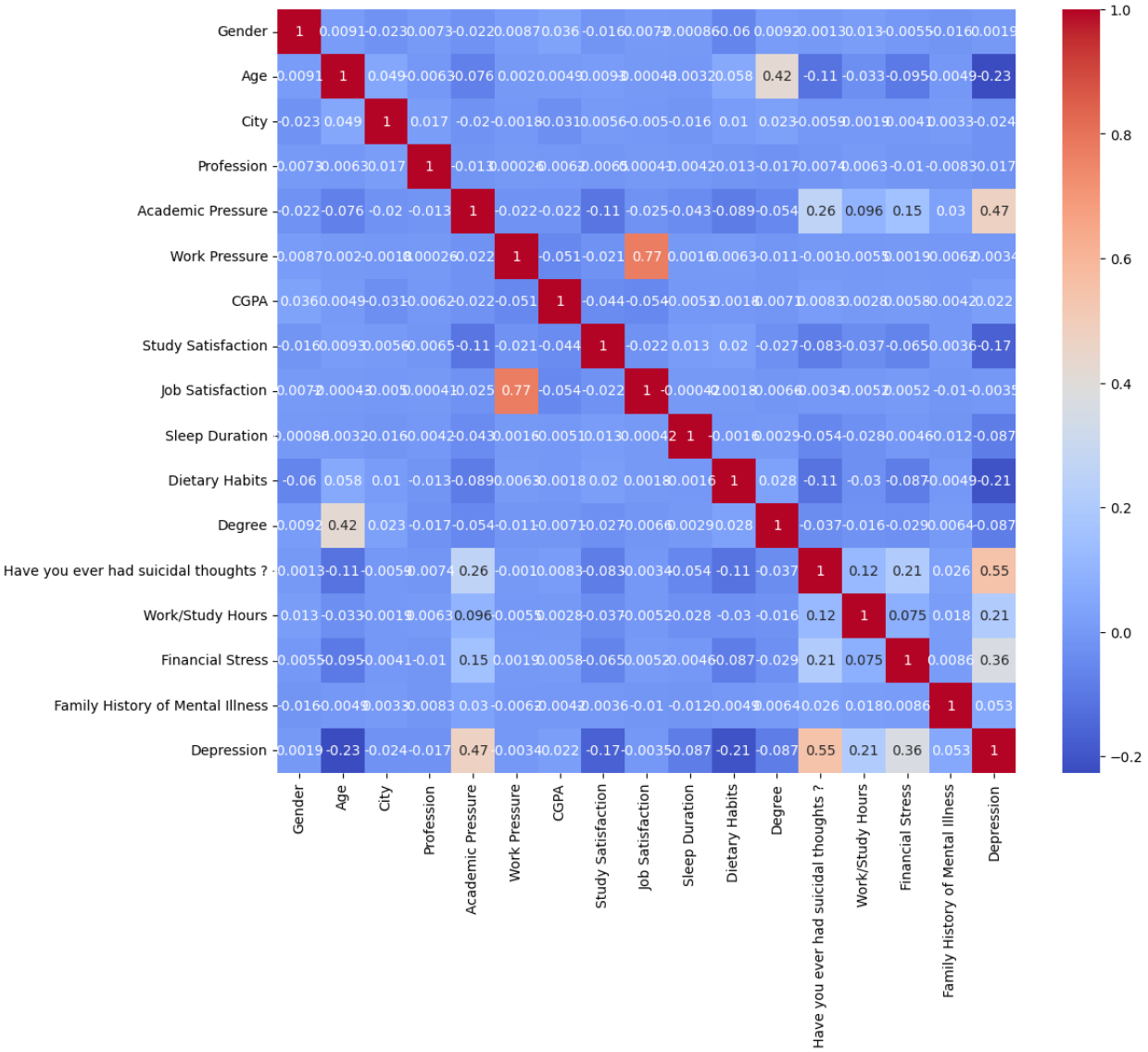

Figure 3.1 Pearson Correlation Matrix for depression

3.2 Classification Reports

Classification reports were generated for each model. These reports show accuracy, precision, recall and F1-scores for both non-depression and depression classes. Performance of all the models was higher for the depression class than for non-depression class.

```
Logistic Regression
--------------------------------------
              precision    recall  f1-score   support

           0       0.82      0.79      0.81      2311
           1       0.86      0.88      0.87      3264

    accuracy                           0.84      5575
   macro avg       0.84      0.84      0.84      5575
weighted avg       0.84      0.84      0.84      5575
```

Figure 3.2 Classification Report for Logistic Regression

As shown in Figure 3.2, the Logistic Regression model had an accuracy score of 84%. It achieved a precision of 86% and recall of 88% for the depression class which resulted in an F1-score of 87%. For the non-depression class, it had a precision of 82% and recall of 79%, resulting in an F1-score of 81%.

```
Naive Bayes
--------------------------------------
              precision    recall  f1-score   support

           0       0.77      0.85      0.81      2311
           1       0.89      0.82      0.85      3264

    accuracy                           0.83      5575
   macro avg       0.83      0.83      0.83      5575
weighted avg       0.84      0.83      0.83      5575
```

Figure 3.3 Classification Report for Naïve Bayes

As seen in Figure 3.3, the Naïve Bayes model had an accuracy score of 83%. It achieved a precision of 89% and recall of 82% for the depression class which resulted in an F1-score of 85%. For the non-depression class, it had a precision of 77% and recall of 85%, resulting in an F1-score of 81%.

```
Random Forest
---------------------------------------
              precision    recall  f1-score   support

           0       0.81      0.79      0.80      2311
           1       0.85      0.87      0.86      3264

    accuracy                           0.84      5575
   macro avg       0.83      0.83      0.83      5575
weighted avg       0.84      0.84      0.84      5575
```

Figure 3.4 Classification Report for Random Forest

As seen in Figure 3.4, the Random Forest model had an accuracy score of 84%. It achieved a precision of 85% and recall of 87% for the depression class which resulted in an F1-score of 86%. For the non-depression class, it had a precision of 81% and recall of 79%, resulting in an F1-score of 80%.

3.3 Performance Evaluation of the Random Forest Model

A confusion matrix was created to detail the correct and incorrect classifications in depression predictions in the Random Forest model as seen in Figure 3.5. The confusion matrix allows for a comparison of the actual classes with the classes predicted by the model. This is critical for analyzing the types of errors encountered.

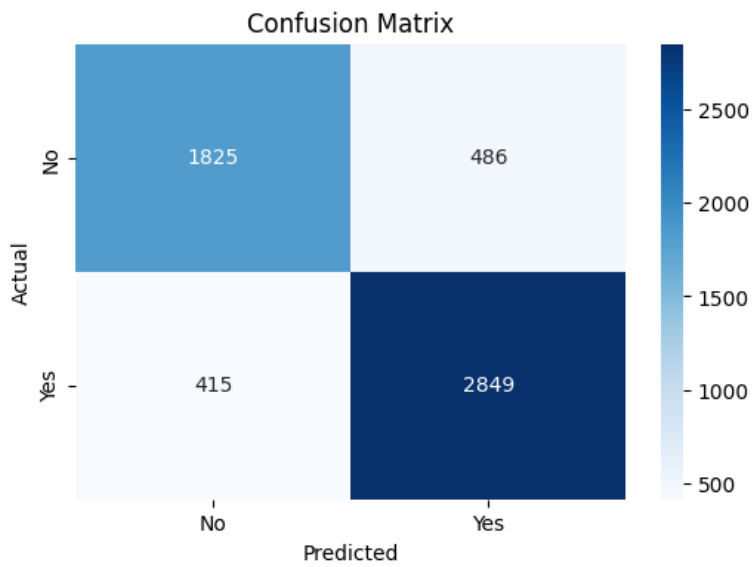

Figure 3.5 Confusion Matrix for Random Forest

1825 people without depression were accurately identified by the model as "No," whereas 486 people were wrongly predicted to be "Yes." When the model was used for individuals with depression, 2849 were correctly classified as "Yes." The 415 individuals incorrectly predicted as "No" are still relatively small compared to the number of correct predictions. The model is successful in identifying individuals who are at risk. It successfully detects most cases of depression while keeping the number of false negatives relatively low.

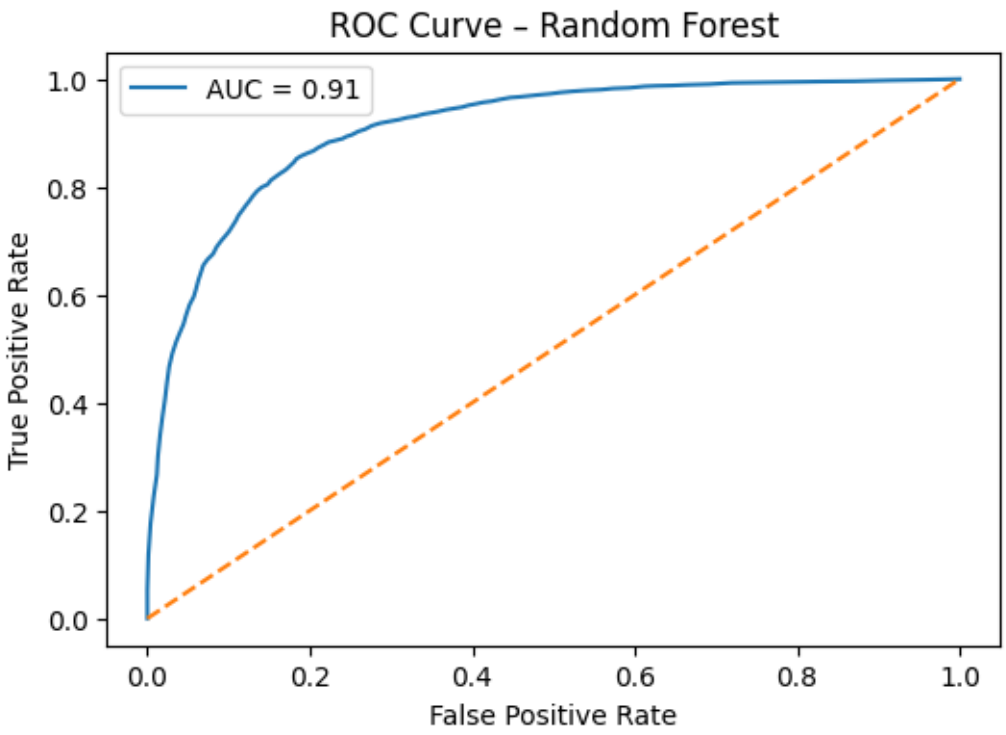

Figure 3.6 ROC Curve for Random Forest

ROC is a performance evaluation tool commonly used in classification studies. The area under the curve, known as the AUC (Area Under the Curve), measures the model's ability to distinguish between individuals with and without depression in the model. A higher AUC value indicates stronger discriminative performance. The high AUC value of 0.91 for the Random Forest model as seen in Figure 3.6, indicates that the model is effective at distinguishing between depressed and non-depressed individuals.

The feature importances obtained from the Random Forest model can be seen in Figure 3.7. These scores show how every feature contributes to the prediction of depression. These values reflect how frequently and effectively the variable is used by the model to reduce classification error across decision trees.

In the resulting table, variables related to psychological distress and stress factors such as suicidal ideation, academic pressure, and financial stress emerge as the most influential features. This indicates that these factors play a significant role in the model's decision-making process. Lower importance scores are seen for lifestyle related factors such as age, gender, dietary habits, and sleep duration. While these variables contribute less individually, they can provide complementary information when combined with other characteristics. This analysis supports the findings of the correlation analysis, highlighting stress and social factors as key drivers of depression prediction.

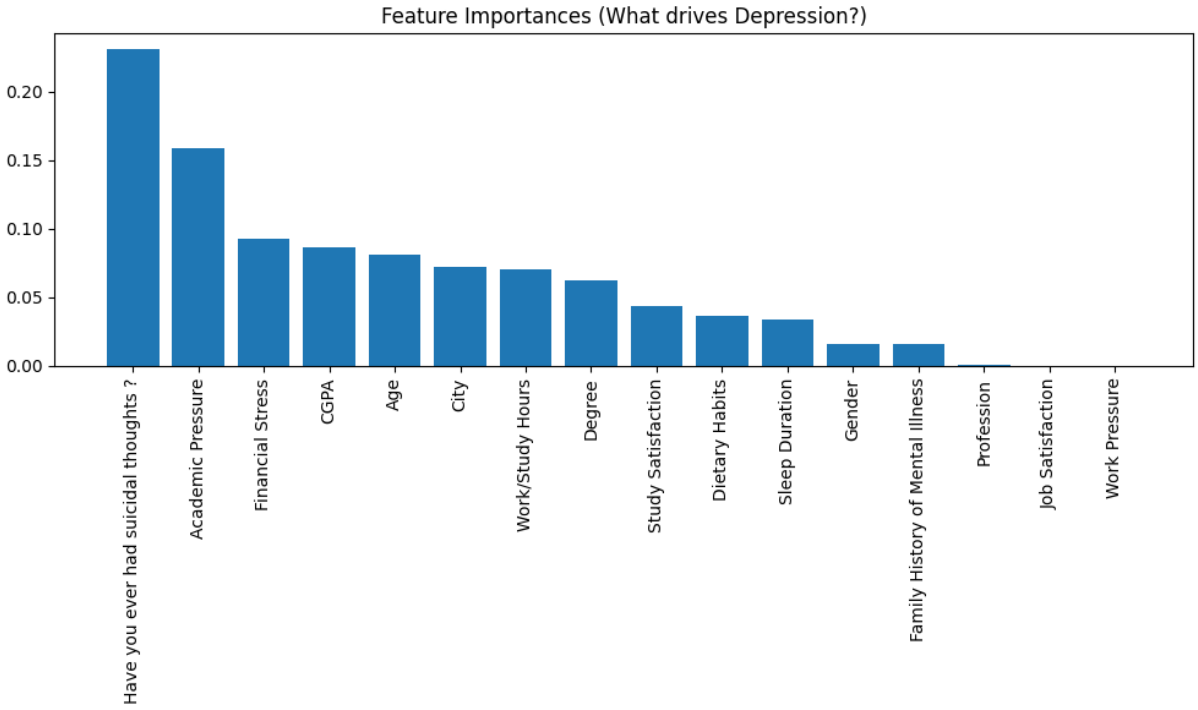

Figure 3.7. Feature Importance Scores from the Random Forest Model

3.4 Descriptive Analysis of Key Predictors

The figures below show the rates of depression in different categories of selected high-impact variables. These graphs visualize the variation in the presence of depression according to variables, compared to a single variable, allowing for further interpretation.

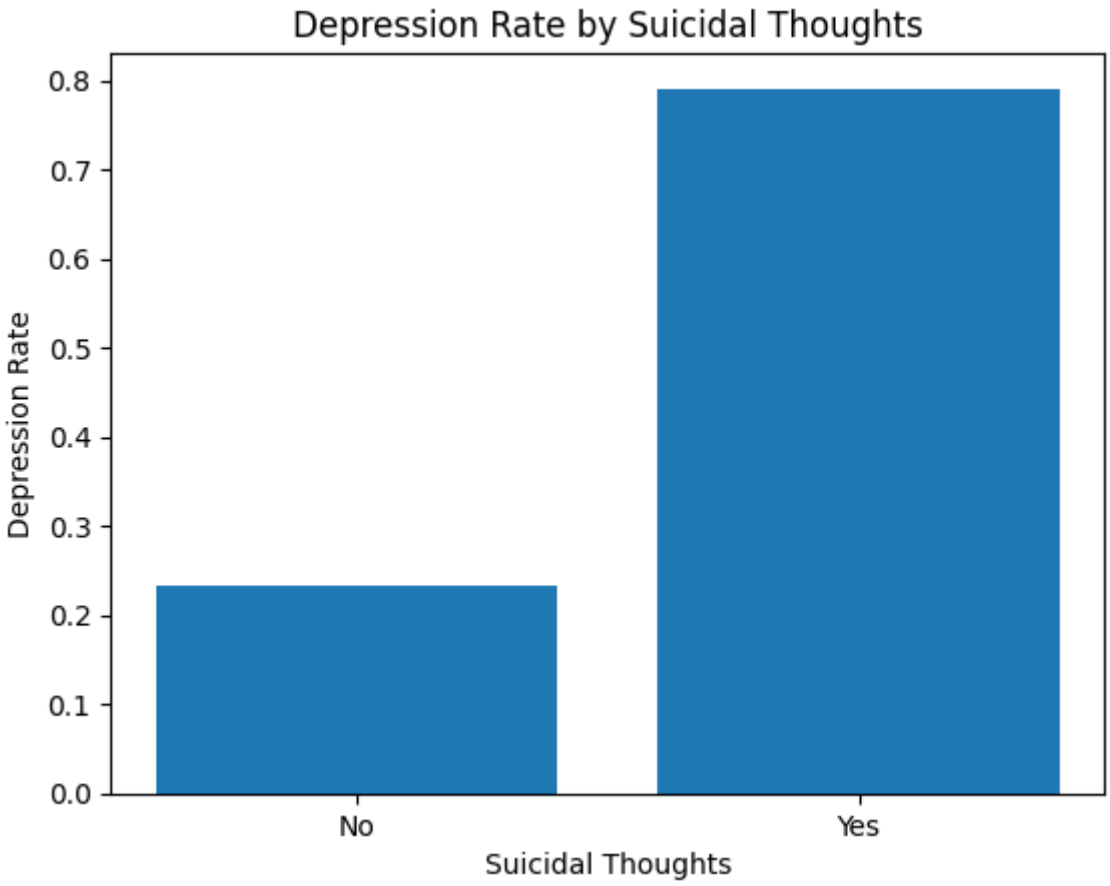

Figure 3.8. Depression Rate by Suicidal Thoughts

As shown in Figure 3.8, the suicidal ideation factor significantly affects depression rates. People with suicidal thoughts have significantly higher rates of depression than those without suicidal thoughts.

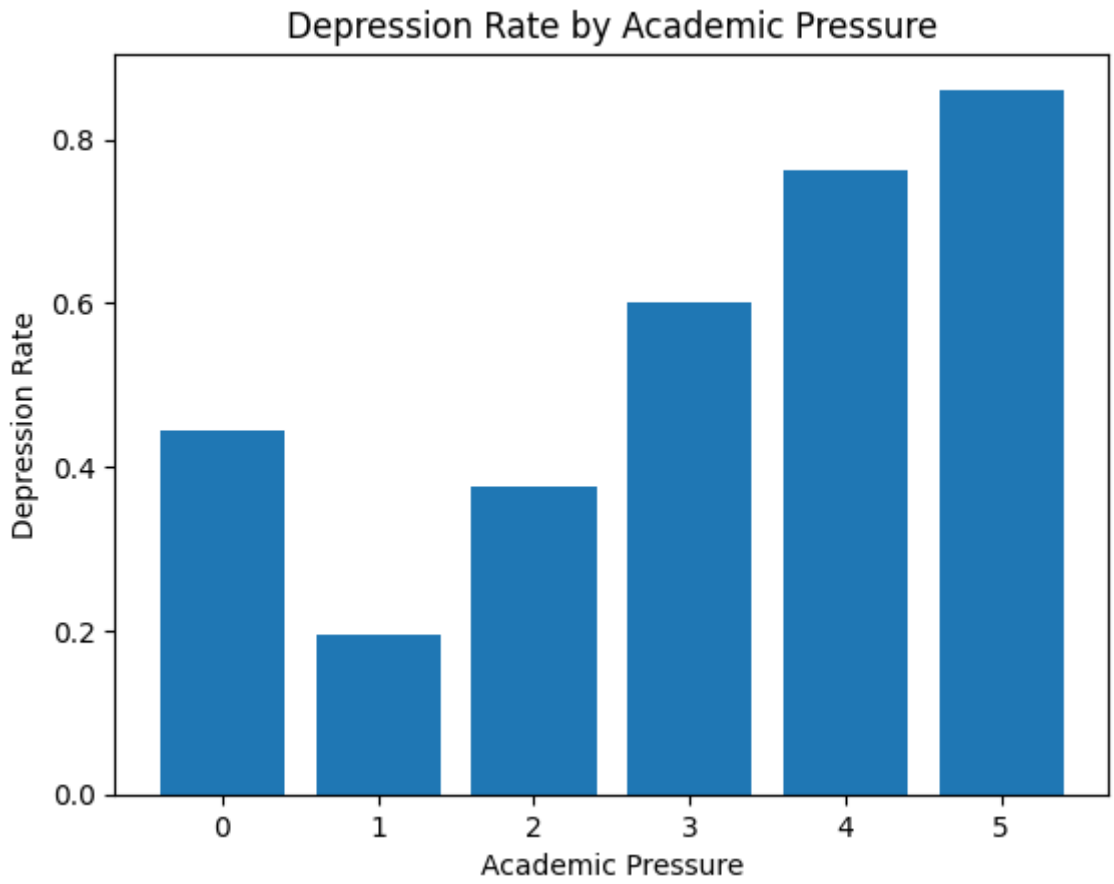

Figure 3.9 Depression Rate by Academic Pressure

Figure 3.9 shows the relationship between academic pressure levels and depression rates graphically. Depression rates generally increase as academic pressure levels rise. While some fluctuations are observed at lower pressure levels, a clear upward trend emerges in higher academic pressure categories. This graph demonstrates a positive relationship between academic pressure and depression. Those experiencing higher academic pressure tend to exhibit higher depression rates when compared to those experiencing lower pressure. It shows that one important stress factor related to depression is academic pressure.

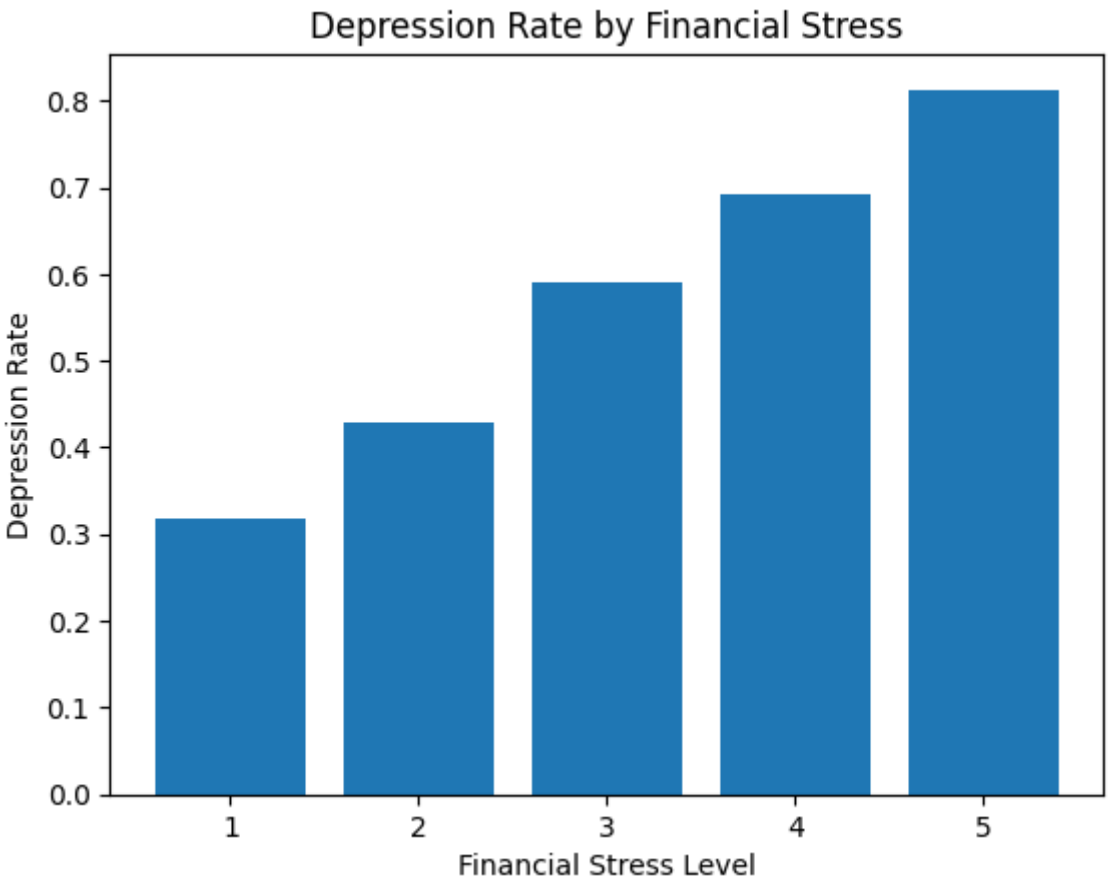

Figure 3.10 Depression Rate by Financial Stress Level

Figure 3.10 helps us understand the relationship between levels of financial stress and rates of depression graphically. Rates of depression increase as levels of financial stress increase. This consistent upward trend suggests a strong positive association between these two variables. This supports the information we obtained from correlation analysis and feature importance.

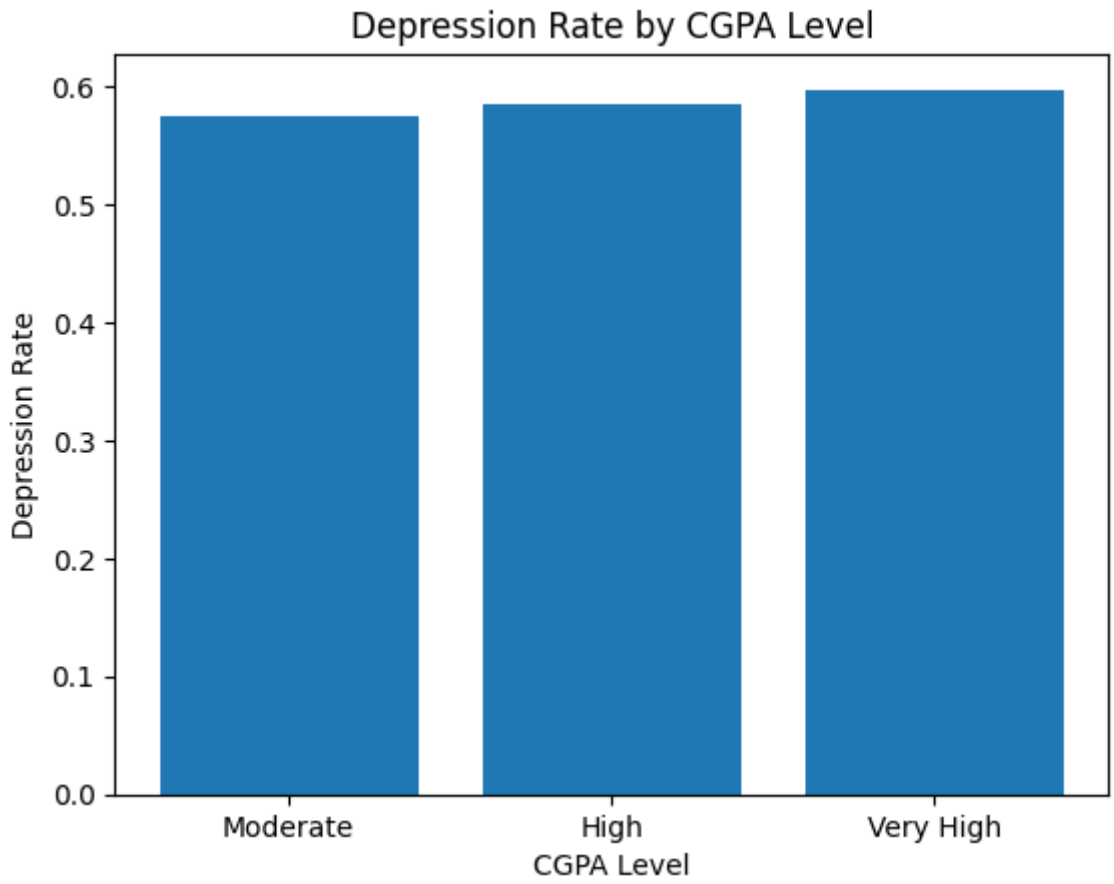

Figure 3.11 Depression Rate by CGPA

Figure 3.11 shows the rates of depression in moderate high and very high CGPAs. As shown in the figure, depression rates remain relatively similar at these levels, with no significant differences observed. The lack of change suggests that academic achievement does not show a correlation with depression. Therefore, CGPA alone does not appear to be a distinguishing factor for depression.

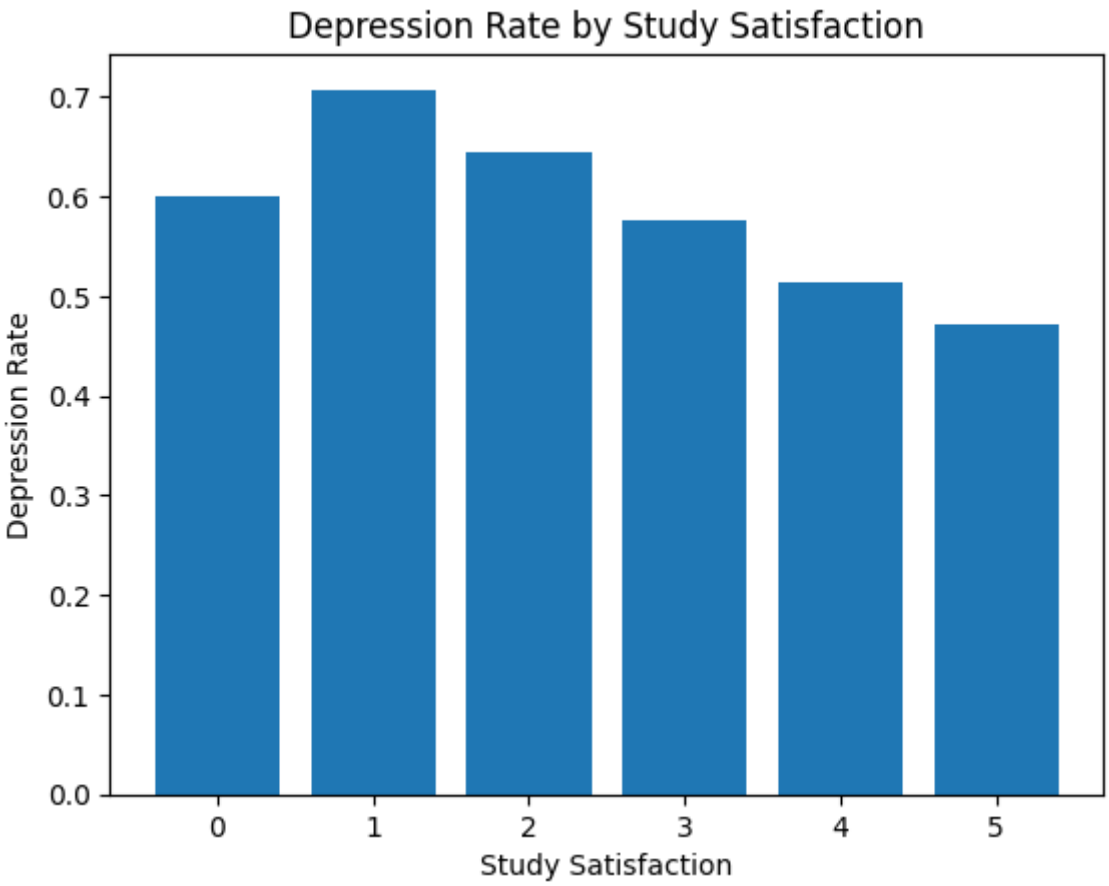

Figure 3.12 Depression Rate by Study Satisfaction

Figure 3.12 shows the relationship between study satisfaction levels and depression rates. The figure shows that as study satisfaction increases, depression rates generally decrease. Individuals with low study satisfaction levels have higher depression rates, while those with high satisfaction levels have relatively lower depression rates. Given the perceived negative correlation, study satisfaction may function as a protective factor against depression.

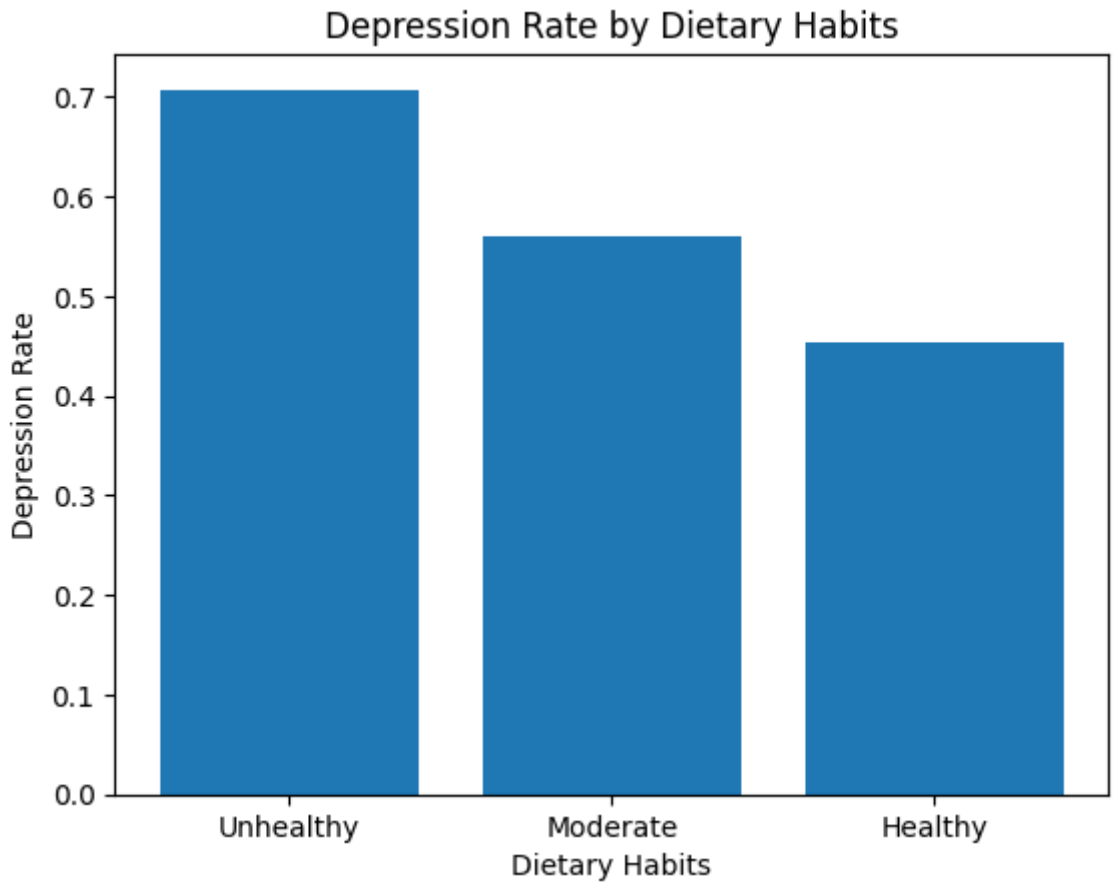

Figure 3.13 Depression Rate by Dietary Habits

Figure 3.13 shows the relationship between different dietary habits and depression rates. The graph shows that depression rates are highest in those with unhealthy dietary habits, while they are lower in those with healthier dietary habits. A decreasing trend in depression rates is observed from unhealthy to healthy dietary habits. Considering this, it can be said that there is a negative relationship between healthy dietary habits and depression. This situation may indicate that healthier dietary habits are associated with lower levels of depression and may function as a protective factor.

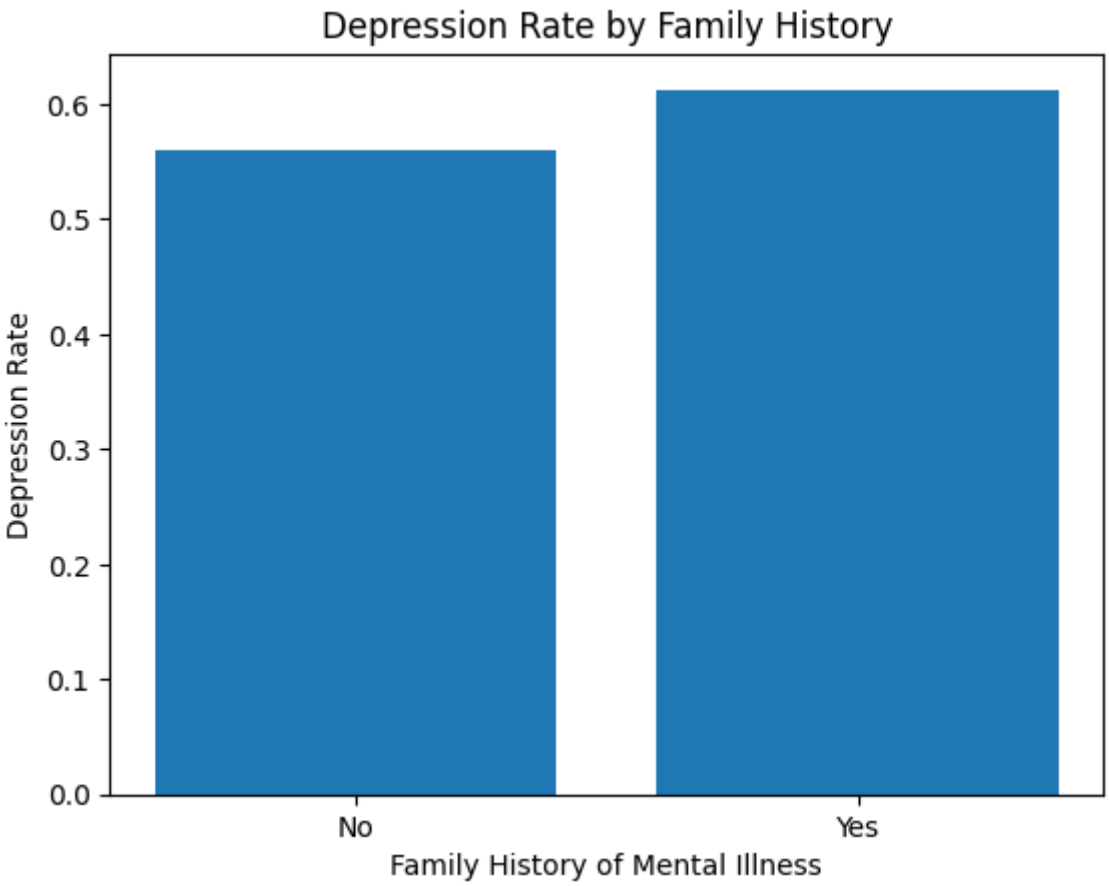

Figure 3.14 Depression Rate Family History of Mental Illness

Figure 3.14 shows the effect of Family History of Mental Illness on the depression rate. It can be said that individuals with a family history of mental illness have a higher depression rate than those without this factor in their family. Although a significant ratio is not seen in the graph, this factor, which is one of the key factors, may gain importance when combined with other factors.

4 DISCUSSION

This project compared the performance of different classification-based machine learning models and analyzed which variables caused depression among students. Logistic Regression, Random Forest and Naïve Bayes models were used, and the accuracy values obtained from these models were very similar. Logistic Regression yielded an accuracy rate of 84%, Random Forest 84%, and Naïve Bayes 83%. This indicates the presence of very strong and regular relationships in the dataset. It also suggests that similar predictions could be obtained with many other models in the same dataset.

While accuracy is important for model selection, it is not sufficient on its own. Therefore, attention was also paid to the ROC-AUC value and modeling capabilities. Due to both its high ROC-AUC value and its ability to model nonlinear relationships, the Random Forest model provides more advantageous and reliable results compared to other models [2]. Since depression is a very complex biological process, analyzing it is equally complex, and therefore observing nonlinear relationships between variables is crucial for accurate analysis. For this reason, the Random Forest model was chosen as the main model.

The fact that all the models used yielded results with approximately 84% accuracy indicates that performance is more related to the structure of the dataset than to the model itself. Since psychological data is based on students expressing their own feelings and thoughts, it is quite normal to find errors and uncertainties in the dataset, and this can make it difficult to determine whether a student is depressed or not [5]. Therefore, even when using more complex machine learning models, a significant increase in success rate has not been observed, indicating that the most realistic level of success achievable with the current variables has been reached.

The significance analysis revealed that work-related variables such as Job Satisfaction, Work Pressure, and Profession had a very low impact and were among the least significant variables in the dataset. This can be explained by demographic characteristics, as the majority of individuals in the dataset were students, and work-related variables had a very limited impact on their lives. The analysis concluded that the primary stress problem for students was not occupational factors, but rather their academic life and lifestyle.

One of the most important findings analyzed in the feature importance analysis is that suicidal ideation is the most dominant variable in predicting depression. While this increases

the accuracy of the model, it also has limitations in terms of early intervention. Suicidal ideation generally increases when depression reaches advanced stages, suggesting that this variable is a characteristic of a consequence rather than a fundamental condition [5]. A model overly dependent on this variable identifies individuals in a serious condition rather than students at risk of depression.

Although genetic predisposition is known to be a significant risk factor for depression, the variable "Family History of Mental Illness" was observed to have a very low significance in the model [5]. This indicates a limitation in the dataset collection method rather than a weakness in biological correlation. Since the dataset used is based on the individual's emotions and thoughts, students may not know or may have provided incorrect information about their family history, as many people feel the need to hide mental illness, especially due to social stigma [6]. The presence of this variable in the dataset caused it to be seen as a noisy and incomplete feature in the model, resulting in lower predictive power compared to variables such as academic pressure or economic status.

The model performance can be significantly improved by adding biological data and data obtained from more objective psychological tests to the dataset. Nevertheless, the data obtained in this project show that machine learning is an effective tool for the early diagnosis of student depression.

5 CONCLUSION

This project analyzed the factors influencing students' depression using a real dataset. Basic data science steps such as data understanding, preprocessing, model development, and analysis were applied. Different classification-based machine learning models were used to analyze whether students were depressed. In addition, the extent to which different values affect depression has also been observed. It has been observed that academic factors are not the sole cause of a student's depression; many other factors such as lifestyle, psychological state, and economic status are also significant contributing factors.

The performance of the models used was quite similar, suggesting that the variables in the dataset could explain depression to a certain extent. The Random Forest model surpassed the other models due to its accuracy rate, ROC-AUC value, and ability to identify nonlinear relationships.

In addition, the main reason for obtaining similar performance values from the models stems more from the dataset than from the model selection. This is because some of the variables in the dataset depend on a person's own feelings and thoughts. This makes it quite difficult to make a clear distinction.

Future studies can improve model performance by using more objective and biological data. In conclusion, this project demonstrates that machine learning allows for a much more detailed analysis of the relationship between students and depression and can significantly contribute to the early diagnosis of at-risk students.

REFERENCES

[1] World Health Organization. (2023). *Depression*. World Health Organization. https://www.who.int/news-room/fact-sheets/detail/depression

[2] Jiang, Y. (2025). Predicting student depression using machine learning: A comparative study. *Highlights in Science, Engineering and Technology*, 12, 45–56.

[3] Journal of Information Systems Engineering and Management. (2024). Machine learning provides powerful tools for prediction in healthcare applications. *Journal of Information Systems Engineering and Management*, 9(2), 1–10.

[4] Shamim, A. (2023). Student Depression Dataset. Kaggle. https://www.kaggle.com/datasets/adilshamim8/student-depression-dataset

[5] Osman, H., Kaya, M., & Demir, A. (2025). Predicting student depression using machine learning: A comparative analysis. *AITU Scientific Research Journal*, 6(1), 22–34.

[6] Thornicroft, G. (2023). Perceived mental illness stigma among family and friends of young people with depression and its role in help-seeking: Qualitative findings on non-disclosure and stigma effects. *BMC Psychiatry*, 22, 107. https://doi.org/10.1186/s12888-022-03754-0